

GhanaMediaWatch Backend Spec

A GPS-gated incident reporting app for Ghana. The camera does not unlock until the device holds an accurate fix; capture always succeeds offline; the queue drains itself when connectivity returns. This is everything the server has to provide for that to work.

Audience backend engineer **Written by** the mobile client team **Status** proposed, not agreed

Base /v1

§ 00

What you're building

Members of the public film incidents — floods, fires, road accidents, illegal mining, power failures — and the app stamps each one with verified coordinates and a timestamp before it ever reaches you. Some reports publish straight to a public feed; some need review first. Institutions subscribe to standing queries over the same crowd-sourced pool.

Four behaviours that shape the API

BEHAVIOUR	WHAT IT MEANS FOR THE SERVER
GPS gate	The camera will not mount until horizontal accuracy reaches 20 m. Every submission therefore arrives with a real fix and an <code>accuracyM</code> . Reports captured through the reduced-accuracy escape hatch are flagged <code>confidence: "low"</code> rather than silently downgraded.
Offline-first	Capture writes to local storage and a SQLite outbox. Uploads happen later, possibly days later, possibly from a different network. Assume every request may be a retry — idempotency is not optional.
Vetting	Reports carry a state. The public feed serves only <code>published</code> . The author sees all of their own, with reasons.
Selective disclosure	A reporter can publish footage while hiding its location, date, or time. This is a privacy guarantee, not a UI preference — see §07.

What the client already handles

So you don't build it twice: local media storage, the SQLite outbox and its state machine, exponential backoff with jitter, background upload scheduling, chunk resumption bookkeeping, connectivity detection, and Wi-Fi-only deferral for video. The server sees the result, not the machinery.

§ 01

Conventions

Authentication

Bearer JWT. Short-lived access token (proposed 15 min), long-lived refresh token (proposed 60 days) held in the device keychain. Three caller identities, and every endpoint below names which it accepts.

IDENTITY	OBTAINED VIA	CAN DO
device	POST /devices	Submit incidents, read the public feed. No profile.
user	POST /auth/signin	Everything above, plus own-report management and interactions.
org_member	POST /auth/signin	Adds the <code>/org/*</code> surface, scoped by role.

Headers on every request

HEADER	NOTES
Authorization	Except <code>POST /devices</code> and <code>/auth/*</code> .
X-Device-Id	The keychain UUID. Sent always , authenticated or not — including on anonymous submissions.
X-Client-Version	App version and build, for support triage.
X-Platform	<code>android</code> or <code>ios</code> .
Idempotency-Key	UUID v4. Required on every state-creating POST.

MUST — IDEMPOTENCY

Capture happens offline and the outbox retries. **The same submission will arrive more than once.** On a repeated `Idempotency-Key` within 24 hours, return the *original response and status code* — not a conflict. Without this, a reporter on a flaky connection publishes the same incident three times to a public feed.

Error envelope

Every non-2xx uses exactly this shape. No bare strings, no HTML error pages.

```
{
  "error": {
    "code": "GPS_ACCURACY_REJECTED",
    "message": "Location accuracy of 84 m exceeds the 20 m limit.",
    "details": { "accuracyM": 84, "limitM": 20 },
    "requestId": "01JBX7Q2K9"
  }
}
```

- `code` — stable SCREAMING_SNAKE_CASE. The client switches on this, never on `message`.
- `message` — developer-facing English. **Never shown to users**; the client maps `code` to a localised string.
- `requestId` — echoed into the client's bug reports. Log it.

Pagination — cursor, not offset

The feed shifts under the reader as reports clear review, so offsets would duplicate and skip. Return `{ items, nextCursor, hasMore }`; `nextCursor` is `null` when exhausted and opaque to the client — encode whatever you like in it. `limit` defaults to 20, caps at 50. Never return a `total`; it is expensive and unused.

Time and coordinates

- Server timestamps are ISO 8601 UTC with `Z`.
- **Device capture time is separate and untrusted.** The client sends `capturedAtIso`, `capturedAtUtcOffsetMinutes`, and `deviceUptimeMs`. Store all three — a device clock inconsistent with its uptime or with server receipt time is a tamper signal for vetting.
- Coordinates: decimal degrees, WGS 84, JSON numbers at 6 dp (~0.11 m). Never strings.
- Distances in metres, durations in ms, sizes in bytes. Integers unless noted.

Domain model

Closed sets. The client holds these as TypeScript unions, so **adding a value is a breaking change** unless the client ships support for it first.

ENUMERATIONS

TS

```
// grouped into colour families in the UI: the hue signals the kind of
// incident, the label says which one
type IncidentCategory =
  | 'fire' | 'accident' | 'disorder'           // emergency
  | 'infrastructure' | 'utility' | 'corruption' // civic
  | 'environment' | 'wildlife'                // environment
  | 'flood' | 'crime' | 'health' | 'other';

type VettingState = 'pending_review' | 'published' | 'rejected' | 'restricted';
type MediaKind = 'photo' | 'video';
type LocationConfidence = 'high' | 'low';
```

Incident — public shape

What the feed and detail endpoints return. Note the nullable metadata fields; that is §07 at work.

```
{
  "id": "inc_01JBX7Q2K9",
  "category": "flood",
  "description": "Culvert blocked, road impassable at the junction.",
  "vettingState": "published",
  "publishedAt": "2026-08-19T14:40:11.000Z",

  "media": {
    "kind": "video",
    "url": "https://cdn.example.gh/m/01JBX7.mp4",
    "posterUrl": "https://cdn.example.gh/m/01JBX7.jpg",
    "width": 1080, "height": 1920,
    "durationMs": 24500, "byteSize": 8421376
  },

  "location": {
    "latitude": 5.603717, "longitude": -0.186964,
    "label": "Kaneshie, Accra", "confidence": "high"
  },

  "capturedAtIso": "2026-08-19T14:31:00.000Z",
  "publisher": { "kind": "anonymous" },
  "counts": { "reactions": 42, "comments": 7 },
  "viewerHasReacted": false
}
```

For an attributed report, `publisher` is

```
{ kind: "user", id, displayName, avatarUrl }.
```

Incident — author's own shape

`GET /me/incidents` returns the above *plus* the fields only the author may see: `rejectionReason`, `displayFlags`, `isAnonymous`, `locationMocked`, and the full precision block (`accuracyM`, `altitude`, `heading`, `speed`).

The client generates a **UUID v4 on first launch** and keeps it in the secure keychain. That is the primary device identifier, sent as `X-Device-Id` on every request. It also sends a secondary platform identifier — Android ID, or iOS `identifierForVendor`.

MUST — WHAT ANONYMITY MEANS HERE

`deviceId` is attached to anonymous submissions too. That is the accountability mechanism the product depends on, and the app tells users so plainly before they publish.

In return, the public API **must never expose** `deviceId`, the author's account, or any derived identifier on an anonymous report — not in the feed, not in detail, not in comments, not in an ETag or a media URL. You retain the link internally for abuse handling and lawful process. **Do not implement anything that would make the app's explainer a lie.**

Both platform identifiers reset on uninstall or factory reset and are **not stable**. Treat them as a weak correlation signal, never a primary key.

POST `/devices`

NO AUTH

Called once on first launch, and again if the token is lost.

REQUEST → 201

JSON

```
{ "deviceId": "9f1c...", "platform": "android",  
  "platformDeviceId": "a1b2c3d4e5f6",  
  "appVersion": "1.0.0", "buildNumber": 42 }  
  
→ { "deviceToken": "eyJ...", "expiresAt": "2026-10-18T...Z" }
```

A 60-second video over Ghanaian mobile data *will* drop mid-flight. Uploads are resumable and chunked, in three phases.

PHASE 1 • INIT

POST /incidents

Metadata only, no bytes.

Returns an `uploadId` and the chunk plan.

PHASE 2 • TRANSFER

PUT

/uploads/{id}/chunks/{n}

Raw bytes. Repeatable and order-independent, because retries reorder.

PHASE 3 • COMMIT

POST

/uploads/{id}/complete

Server assembles, verifies the hash, enqueues for vetting.

If the app dies mid-upload, the client calls `GET /uploads/{id}` on next launch to learn which chunks you already hold, then resumes. **It does not restart from zero.**

POST

/incidents

DEVICE • USER

Metadata only. Requires `Idempotency-Key`.

REQUEST BODY

JSON

```
{
  "clientId": "e3a1...",           // the client's local row id
  "category": "flood",
  "description": "Culvert blocked, road impassable.",
  "isAnonymous": true,
  "displayFlags": { "showLocation": true,
                    "showDate": true,
                    "showTime": false },

  "location": {
    "latitude": 5.603717, "longitude": -0.186964,
    "accuracyM": 8.4, "altitude": 61.2,
    "heading": 142.0, "speed": 0.0,
    "confidence": "high", "isMocked": false
  },

  "capturedAtIso": "2026-08-19T14:31:00.000Z",
  "capturedAtUtcOffsetMinutes": 0,
  "deviceUptimeMs": 184523110,     // tamper signal

  "media": {
    "kind": "video", "mimeType": "video/mp4",
    "byteSize": 8421376, "durationMs": 24500,
    "width": 1080, "height": 1920,
    "sha256": "9f86d081884c7d65..."
  }
}
```

RESPONSE • 201

JSON

```
{ "incidentId": "inc_01JBX7Q2K9",
  "uploadId": "upl_01JBX7Q2K9",
  "chunkSizeBytes": 5242880,
  "chunkCount": 2,
  "expiresAt": "2026-08-20T14:31:00.000Z" }
```

The client proposes 5 MiB chunks, but **your response is authoritative** — it will use whatever `chunkSizeBytes` you return. Echo `clientId` back on the resulting incident so the outbox reconciles without relying on ordering.

PUT `/uploads/{uploadId}/chunks/{index}`

DEVICE • USER

`index` is **0-based**. Body is `application/octet-stream` — raw bytes, no multipart wrapper. Returns `{ index, received, bytesReceived }`.

Must be idempotent. Re-sending a chunk you already hold returns `200`, not an error.

POST `/uploads/{uploadId}/complete`

DEVICE • USER

Empty body. Returns the full incident in its author shape, with `vettingState` set by your routing rules.

Verify `sha256` here and reject with `MEDIA_HASH_MISMATCH` on a mismatch — that is the client's signal to re-upload rather than silently publishing corrupt footage.

Only after this succeeds does the client delete its local media file. It keeps the metadata row so the reporter retains a personal history — which is why media retention is [an open question](#).

GET `/uploads/{uploadId}`

DEVICE • USER

Resume support. Returns `receivedChunks: [0]` so the client sends only what is missing. If the window has elapsed, return `410 UPLOAD_EXPIRED` and the client restarts from `POST /incidents`.

GET

/incidents

DEVICE • USER

The public feed. **Returns only** `published` . Never leak another state here.

PARAM	NOTES
limit, cursor	Cursor pagination, see §01.
category	Repeatable; multiple values mean OR.
near	<code>lat,lng</code> — enables <code>distanceM</code> on each item.
radiusM	Requires <code>near</code> .
since, until	ISO 8601, applied to <code>publishedAt</code> .
sort	<code>recent</code> (default) or <code>nearby</code> .

GET

/incidents/map

DEVICE • USER

The map tab needs clusters, not a page of full records. `?bbox=-0.3,5.5,-0.1,5.7&zoom=12` returns `{ clusters: [{ latitude, longitude, count, categories }], markers: [...] } .`

Cluster server-side. Returning 4,000 raw points and clustering on-device will not hold 60 fps on a mid-range Android phone.

GET

/incidents/{id}

DEVICE • USER

Public shape. If it is not published and the caller is not the author, return `404 INCIDENT_NOT_FOUND` — **not** `403` , which would confirm the report exists.

GET

/me/incidents

USER

All states, author shape, with `rejectionReason` where applicable. Filter with `?state=pending_review` .

PATCH

`/incidents/{id}`

USER • AUTHOR ONLY

The account owner can change `displayFlags` *after* publication — that is a product requirement, not an edge case.

Also accepts `description`. **Editing the description after publication must re-enter review**, otherwise it is a trivial bait-and-switch on approved footage.

DELETE

`/incidents/{id}`

USER • AUTHOR ONLY

Soft-delete. Returns `204`.

§ 06

Vetting

Every report enters at `pending_review` and leaves to exactly one terminal state.

STATE	IN FEED?	AUTHOR SEES?	MEANING
<code>pending_review</code>	No	Yes, with a banner	Awaiting a human or automated decision.
<code>published</code>	Yes	Yes	Cleared for public display.
<code>rejected</code>	No	Yes, with the reason	Declined. <code>rejectionReason</code> is shown to the reporter verbatim — write it for them, not for staff.
<code>restricted</code>	No	Yes, with a banner	Held for security sensitivity. Visible to authorised org accounts only.

The brief allows reports to be "vetted or automatically uploaded ... based on public or security sensitivity". **The client does not decide this** — it renders whatever state you return. Which categories auto-publish, and who triggers human review, is [question 3](#).

Display flags

Read this section twice. Its failure mode is exposing a reporter's location.

Each report carries three booleans, set by the reporter at review time and editable later: `showLocation`, `showDate`, `showTime`. They are **display flags, not storage flags** — you still store the true coordinates and capture timestamp for every report, always, because vetting, abuse investigation, and lawful requests all depend on it.

MUST – SUPPRESS AT THE SOURCE

On any endpoint serving a *public* audience — feed, map, public detail, comments — the server must **omit or null the suppressed fields**. Do not send them and trust the client to hide them.

WHEN FALSE	PUBLIC RESPONSES MUST...
<code>showLocation</code>	Null <code>latitude</code> , <code>longitude</code> , and <code>label</code> . Also drop the item from <code>/incidents/map</code> and from <code>near / radiusM</code> filtering — a marker on a map is a location disclosure.
<code>showDate</code>	Null <code>capturedAtIso</code> . Use <code>publishedAt</code> truncated to the hour if you need an ordering key.
<code>showTime</code>	Truncate <code>capturedAtIso</code> to midnight UTC.

The client enforces the identical rules independently, with unit tests asserting a hidden value never renders. **Both layers are required.** Client-side hiding alone means the true coordinates travel to every phone that scrolls past the report and sit in an inspectable cache. Server-side alone means one client bug leaks them. Defence in depth is not negotiable here.

If `showDate` is false but `showTime` is true, suppress both — a bare time with no date is meaningless and still narrows the window. The client greys out that combination, but do not rely on it.

Org tier

Institutions and businesses subscribe to standing queries over the crowd-sourced pool.

MUST – AUTHORISATION

The client hides `/org` routes from non-members as a courtesy. **That is not *authorisation***. Every endpoint here must independently enforce membership and role. Roles: `owner` , `admin` , `analyst` , `viewer` .

`GET``/org/dashboard``ORG_MEMBER`

Counts by category and by state, a daily trend series for the sparkline, and recent high-priority reports.

GET

POST

`/org/queries`

ORG_MEMBER

An org's standing watch criteria — categories, keywords, a geofence, a time window, and alert preferences.

SAVED QUERY

JSON

```
{
  "name": "Flooding - Odaw catchment",
  "categories": ["flood", "infrastructure"],
  "keywords": ["culvert", "drain"],
  "geofence": {
    "kind": "circle",          // or "polygon" + points[]
    "center": { "latitude": 5.603717, "longitude": -0.186964 },
    "radiusM": 3000
  },
  "timeWindow": { "since": "2026-08-01T00:00:00.000Z" },
  "alerts": { "push": true, "email": false }
}
```

A saved query must never match a report whose `showLocation` is false via its geofence. Suppressed location means suppressed from location-based matching — including for paying customers.

POST

`/org/incidents/{id}/status · /notes`

ORG_MEMBER

Inbox triage: `in_review`, `actioned`, `dismissed`; plus internal notes that are **never public**.

POST

`/org/exports`

ORG_MEMBER

Returns `202` with an `exportId`. The client polls `GET /org/exports/{id}` until ready, then hands the URL to the OS share sheet.

Async, not a blocking download — a 90-day CSV will time out a mobile request.

Error codes

HTTP	CODE	CLIENT BEHAVIOUR
400	VALIDATION_FAILED	Shows a field error. Does not retry.
401	TOKEN_EXPIRED	Refreshes once, replays the request.
401	TOKEN_INVALID	Signs the user out.
403	FORBIDDEN	Shows an explanatory screen.
404	INCIDENT_NOT_FOUND	Generic not-found.
409	IDEMPOTENCY_CONFLICT	Logs; does not retry.
410	UPLOAD_EXPIRED	Restarts from <code>POST /incidents</code> .
413	MEDIA_TOO_LARGE	Fails permanently, tells the user.
422	GPS_ACCURACY_REJECTED	Fails permanently with an explanation.
422	MEDIA_HASH_MISMATCH	Re-uploads from chunk 0.
429	RATE_LIMITED	Backs off, honours <code>Retry-After</code> .
500	INTERNAL	Retries with backoff.
503	MAINTENANCE	Retries with backoff; shows a banner.

MUST – RETRYABLE VS TERMINAL

The outbox retries `429`, `5xx`, and network failures *forever* with backoff (2s → 8s → 30s → 2m → 10m → 1h). It fails an item permanently on `400`, `403`, `413`, and `422`.

Returning `500` for what is really a validation error means every affected phone retries a doomed upload, hourly, indefinitely.

Open questions

None of these are decided. The client picked a default to stay unblocked. Each is cheap to change now and expensive later — **these are the answers we need from you.**

#	QUESTION	CLIENT ASSUMES	WHY IT MATTERS
1	Media retention — how long is footage kept?	Indefinite	The client deletes its local copy once you confirm completion. Expire media at 90 days and the reporter's own history breaks.
2	Max file size	200 MB / 60 s	Drives the recording cap and whether we transcode on-device.
3	Who triggers review; which categories auto-publish?	Everything enters review	If <code>fire</code> and <code>accident</code> auto-publish for speed, the submit confirmation must say so.
4	Do comments exist in v1?	No — the count is in the shape, no UI ships	Adding them later is additive; the action rail has the slot reserved.
5	Org seat / pricing model	Unlimited seats, no in-app billing	Capped or purchasable seats change the members screen and pull in app-store billing.
6	Push payload shape	<code>{ incidentId, queryId, title, body }</code>	The client routes the notification tap by <code>incidentId</code> .
7	Consent copy before publishing footage of people	Placeholder text	This needs a lawyer, not either of us. Ghana's Data Protection Act 2012 (Act 843) applies to publishing identifiable people. The screen exists; the words must be supplied.

#	QUESTION	CLIENT ASSUMES	WHY IT MATTERS
8	Server-side GPS accuracy limit	Accept and flag <code>confidence: "low"</code>	Do you reject loose fixes outright, or accept and flag?
9	Mocked-location reports	Accepted, flagged <code>isMocked</code>	Confirm you want them stored at all.
10	Account deletion	<code>DELETE /me</code> cascades and anonymises prior reports	Interacts with retention and with law-enforcement obligations.

The mobile client runs today against a `MockApiClient` implementing every endpoint above with seeded fixtures, so this contract is already exercised end to end. Where your implementation needs to differ, say so early — the client's API layer is a single swappable interface, and changing it before the screens are built costs almost nothing.